

Rajvrat Thapliyal

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EDUCATION

National University of Singapore - <i>BSc (Hons) Data Science and Analytics</i>	Aug 23' – Present
• Special Program in Science (SPS) multidisciplinary research program	
Stanford University - <i>Management Science & Engineering (NUS Overseas Entrepreneurship Program)</i>	Aug 25' – Dec 25'

WORK EXPERIENCE

Product Manager Intern - AI Applications, Workato Inc (Palo Alto)	Aug 25' – Present
- Built and owned the retrieval pipeline and evaluation framework for Workato GO, an AI-powered enterprise workflow automation platform; improved search quality by 23% through graph database-backed retrieval and operationalized NDCG as the core metric, presenting findings to the Director of Product. - Deployed automated daily evaluation pipelines on EC2 with regression-aware quality gates and structured source provenance tracking, directly informing feature rollout decisions across the full product stack. - Led GO's MCP client product strategy, authoring full PRDs and designing the connections page architecture; conducted structured competitive test matrices against leading AI platforms, shaping engineering priorities and shipping decisions. - Leading the complete upgrade of Workato GO's model infrastructure to Anthropic Claude; designed a surrogate routing system benchmarking Claude Sonnet/Opus/Haiku against GPT-4.1 across tool-calling, long-context coherence, and agentic task performance. - Took ownership of GO's bug fixes and feature enhancements, coordinating across UI, engineering, security, and documentation teams as the primary point of contact for customer-facing issues end-to-end.	
ML Researcher, Satellite Technology and Research (STAR) Centre	Jan 25' – May 25'
- Redesigned physics-informed neural networks for large-scale satellite trajectory propagation across 8000+ Starlink satellites using a 90GB orbital dataset, reducing orbital deviation error by up to 81% across regimes. - Refactored and parallelized a satellite trajectory propagator originally trained on A100 cluster infrastructure, reducing end-to-end training time from 71 days to 13 minutes on a personal GPU via satellite-specific lightweight models. - Led architecture design, ablation studies, and error-envelope validation; awarded Best Research Project at NUS SPS Showcase; manuscript currently in preparation for publication.	
ML Engineer, OnSchool EdTech Group	May 24' – Jul 24'
- Built XGBoost and Prophet ensemble forecasting models and designed a constrained budget allocation optimizer - Increased L1 customer acquisition by 73% within one month by optimizing multi-channel marketing spend.	

PROJECTS

StarGuard - Autonomous Orbital Defense (Treehacks '26 NVIDIA Track Finalist)	Feb 26'
- Fine-tuned two YOLOv8 models on synthetic star-tracker images for satellite detection in low-light orbital conditions. - Implemented custom Unscented Kalman Filter for trajectory estimation; integrated convex optimization for evasive manoeuvre planning. - Integrated vision-language model for contextual adversarial satellite risk assessment; GPU-parallelized orbital simulations - Invited by NVIDIA's Physical AI team to demo the project for GTC 2026.	
SiteOps - AI Construction Safety Platform (HackHarvard Winner)	Nov 25'
- Built real-time computer vision pipeline for worker proximity monitoring and fall detection using object detection models with spatial distance estimation between personnel and heavy machinery. - Designed event inference system to translate model outputs into structured safety signals, detecting incidents such as unsafe proximity, fallen workers, and overcrowding for automated alerting and workflow integration.	

SKILLS

- Machine Learning & Modeling: PyTorch, YOLOv8, XGBoost, Optuna, Hugging Face
- LLMs & GenAI: OpenAI API, LangChain, RAG pipelines, vector databases, agent orchestration frameworks
- Cloud & Infrastructure: AWS EC2/S3, REST APIs, Linux, Snowflake
- Programming: Python (primary), SQL, Java, TypeScript
- Scientific Computing: NumPy, SciPy, pandas, scikit-learn
- Product & Business: Product Management, PRD Authoring, Cross-functional Team Management, Microsoft Suite, Presentation & Public Speaking, Tableau